

6. You are exposed to natural sources of background radiation all the time.
The level of background radiation varies depending on where you live.

(a) (i) Name **one** cause of background radiation.

[1]

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(ii) Give a reason why the level of background radiation depends on where you live.

[1]

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- (b) Having an X-ray taken of part of the body exposes the person to ionising radiation.
The following table gives the effect on the body from X-rays.

X-ray	Radiation's effect on the body	
	Radiation dose (units)	Equivalent number of hours of background radiation
chest	100	200
hand	1	2
dental	5	10

- (i) State which X-ray gives the most risk to the patient and give a reason for your answer.

[2]

X-ray

Reason

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- (ii) A person has 1 chest X-ray and 4 dental X-rays in his lifetime.
Toni suggests that this is equivalent to 240 hours of extra background radiation.
Use data from the table to decide whether you agree with this suggestion.
Space for calculation.

[3]

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